Mathematics practice booklet: Grade 0 expectations

KG2 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 0 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: K.OA.A.1, K.OA.A.2, K.OA.A.3, K.OA.A.4, K.OA.A.5

A worked example

Put 3 dates on a plate. Then put 2 more dates on the plate. How many dates are on the plate now?

1. Count the first dates: 1, 2, 3.
2. Add the 2 new dates and keep counting: 4, 5.
3. Putting 3 and 2 together makes 5.

5 dates

Questions

1. Maryam has 3 buttons. Her friend gives her 2 more buttons. How many buttons does Maryam have now?

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2. There are 4 cups on the table. Salem puts 1 more cup on the table. How many cups are there?

   A. 3

   B. 5

   C. 6

3. Hamdan has 5 stones. He gives 2 stones to his brother. How many stones does Hamdan have left?

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4. Noura has 2 red cups and 4 blue cups. How many cups does she have in all?

   A. 8

   B. 2

   C. 6

5. There are 7 dates in a bowl. Layla eats 3 of them. How many dates are left in the bowl?

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6. Take 5 buttons. Show 2 ways to split them into two small groups. What are the groups?

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7. Khalid holds up 3 fingers. How many more fingers make 5?

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8. Omar has 6 stones. How many more stones does he need to make 10?

   A. 4

   B. 5

   C. 16

9. There are 10 cookies on a tray. Hind takes 4 cookies. How many cookies are left on the tray?

   A. 14

   B. 5

   C. 6

10. Look at the pattern: red button, blue button, red button, blue button, red button. Which button comes next?

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11. Look at the pattern: date, date, stone, date, date, stone, date, date. What comes next? How do you know?

Section 2: Shapes, measuring and space

Standards practised: K.G.A.1, K.G.A.2, K.G.A.3, K.G.B.4, K.MD.A.1, K.MD.A.2

A worked example

Which is longer: a new pencil or a small eraser? Put them side by side with the ends lined up at one end.

1. Line up the left ends of both things.
2. Look at the other ends. The one that sticks out more is longer.
3. Say the answer: the pencil reaches further, so it is longer.

The pencil is longer, because it reaches further

Questions

1. Maryam puts a round plate on the table. What shape is the plate?

   A. Square

   B. Circle

   C. Triangle

2. A cracker has 4 sides that are all the same length. What shape is it?

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3. Salem draws a triangle. How many sides does it have?

   A. 4

   B. 2

   C. 3

4. Noura has a rectangle tray. How many corners does the tray have?

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5. Hamdan kicks a ball. Is a ball a flat shape or a solid shape? What is its name?

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6. A tin of chickpeas is a solid shape. Which solid shape is it?

   A. Cylinder

   B. Cube

   C. Sphere

7. A ball is under the chair. Is the ball above the seat or below the seat?

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8. Maryam stands between Salem and Noura. Who is in the middle?

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9. Layla looks at a pencil and a date palm tree. Which is longer?

   A. A pencil

   B. A date palm tree

   C. A date

10. Hind lifts a watermelon and a date. Which is heavier?

    A. A watermelon

    B. A date

    C. They weigh the same

11. Omar's rope is longer than Layla's rope. Which rope is shorter?

Section 3: Data, graphs and chance

Standards practised: K.MD.B.3, K.CC.C.6, K.CC.C.7, K.CC.B.5

A worked example

On a tray there are 3 dates and 2 stones. Sort them into two groups. Which group has more?

1. Make one group of dates and one group of stones.
2. Count each group: dates 1, 2, 3 and stones 1, 2.
3. 3 is more than 2, so the dates are more.

The dates have more

Questions

1. Sort these into groups: 3 dates, 2 stones and 1 button. How many dates are there?

   _____

2. There are 4 red cups and 3 blue cups. How many cups are there in all?

   A. 8

   B. 7

   C. 1

3. Salem has 5 stones and Hind has 3 stones. Who has more stones?

   _____

4. Maryam has 4 buttons and Omar has 6 buttons. Who has fewer buttons?

   _____

5. There are 4 dates and 4 cookies on a plate. Which sentence is true?

   A. There are more dates

   B. There are more cookies

   C. There is the same number of each

6. Sort the spoons and forks. There are 3 spoons and 4 forks. How many are there in all?

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7. There are 5 cups and 3 plates. How many more cups than plates are there?

   A. 2

   B. 3

   C. 8

8. There are 2 big stones and 5 small stones. Which group has fewer?

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9. There are 8 toys. 5 are cars and the rest are balls. How many balls are there?

   A. 13

   B. 3

   C. 4

10. 5 children want to sit down and there are 4 chairs. Is there a chair for every child? How many more chairs are needed?

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11. Sort 10 stones: 6 are smooth and 4 are rough. Which group has more, and how many more?

Section 4: Numbers and calculating

Standards practised: K.CC.A.1, K.CC.A.2, K.CC.A.3, K.CC.B.4, K.CC.B.5, K.CC.C.6, K.CC.C.7, K.NBT.A.1

A worked example

Count the dates in the bowl: there are 13. Show 13 as ten and some ones.

1. Make a group of 10 dates.
2. Count the dates left over: 1, 2, 3.
3. 13 is 10 and 3 ones.

13 is ten and 3 ones

Questions

1. Count out loud from 1 to 10. What number comes after 7?

   _____

2. Maryam counts 12 dates one by one. When she says the last number, how many dates are there?

   _____

3. Salem has 9 stones and finds 1 more. What is one more than 9?

   A. 10

   B. 8

   C. 11

4. Noura has 6 cups and puts 1 away. What is one less than 6?

   A. 7

   B. 4

   C. 5

5. Start at 8 and count on. Say the next three numbers.

   _____

6. Count by tens: 10, 20, 30. What number comes next?

   A. 31

   B. 40

   C. 50

7. Which one is the number fifteen written with digits?

   A. 50

   B. 51

   C. 15

8. Salem has 8 stones and Noura has 6 stones. Which number is greater?

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9. Hamdan has 9 dates and Layla has 7 dates. Which number is smaller?

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10. 13 is ten and how many ones?

    A. 3

    B. 1

    C. 13

11. Ten buttons and 4 more buttons make what number? Which number comes just before 20?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. Count 3, then 4, 5. So Maryam has 5 buttons now.
2. B. 5. 4 and 1 more is 5, because one more is the next number. If your child chose C (6), they added 2 instead of 1.
3. Start at 5 and take away 2: 4, 3. So 3 stones are left.
4. C. 6. Put 2 and 4 together: 2, then 3, 4, 5, 6. If your child chose B (2), they counted only the red cups.
5. Take away 3 from 7 by counting back: 6, 5, 4. So 4 dates are left.
6. 1 and 4 make 5, and 2 and 3 make 5. Add each pair to check.
7. Hold up 3 fingers, then count on to 5: 4, 5. That is 2 more fingers.
8. A. 4. Count on from 6 to 10: 7, 8, 9, 10. That is 4 more. If your child chose C (16), they added 6 and 10.
9. C. 6. Take away 4 from 10 by counting back: 9, 8, 7, 6. If your child chose A (14), they added instead of taking away.
10. The pattern goes red, blue, red, blue, red. So a blue button comes next.
11. The pattern is date, date, stone, and it repeats. After two dates comes a stone.

Section 2: Shapes, measuring and space

1. B. Circle. A round plate has no corners and no straight sides, so it is a circle. If your child chose A (square), they may have thought of a shape with corners.
2. A square has 4 sides that are the same length. Count them: 1, 2, 3, 4.
3. C. 3. Count the sides of a triangle: 1, 2, 3. If your child chose A (4), they may have thought of a square.
4. A rectangle has 4 corners. Touch each corner and count: 1, 2, 3, 4.
5. A ball is round all over and you can hold it. So it is a solid shape called a sphere.
6. A. Cylinder. A tin has two flat round ends and a curved side, so it is a cylinder. If your child chose C (sphere), they saw it is round but missed the flat ends.
7. Below means lower than. The ball is under the chair, so it is below the seat.
8. Between means in the middle of two. Maryam stands in the middle of Salem and Noura.
9. B. A date palm tree. A palm tree is much taller than a pencil, so it is longer. If your child chose A (a pencil), they may have picked the small thing they can hold.
10. A. A watermelon. is big and a date is tiny, so the watermelon is heavier. If your child chose C (the same), they may not have compared the sizes.
11. Omar's rope is longer, so Layla's rope is the shorter one.

Section 3: Data, graphs and chance

1. Count only the dates: 1, 2, 3. So there are 3 dates.
2. B. 7. Put 4 and 3 together: 4, then 5, 6, 7. If your child chose C (1), they took 3 away from 4 instead.
3. Compare 5 and 3: 5 is more than 3. So Salem has more stones.
4. Compare 4 and 6: 4 is less than 6. So Maryam has fewer buttons.

5. C. There is the same number of each. 4 and 4 are the same number, so neither group has more. If your child chose A (more dates), they may have picked the first group they saw.
6. Put 3 spoons and 4 forks together: 3, then 4, 5, 6, 7. That is 7 things.
7. A. 2. Match 3 cups with 3 plates, and 2 cups are left over. If your child chose B (3), they gave the number of plates.
8. Compare 2 and 5: 2 is fewer than 5. So the big stones are the smaller group.
9. B. 3. Take away the 5 cars from 8 toys: 7, 6, 5, 4, 3. If your child chose A (13), they added instead of taking away.
10. 4 children sit on 4 chairs and 1 child has no chair. So no, 1 more chair is needed.
11. 6 is more than 4, and 6 take away 4 is 2. So the smooth stones have 2 more.

Section 4: Numbers and calculating

1. Count 1, 2, 3, 4, 5, 6, 7, 8. The number after 7 is 8.
2. The last number you say tells how many. So there are 12 dates.
3. A. 10. One more than 9 is the next number when you count, which is 10. If your child chose B (8), they went back instead of forward.
4. C. 5. One less than 6 is the number just before 6, which is 5. If your child chose A (7), they went forward instead of back.
5. Start at 8 and count on: 9, 10, 11. Those are the next three numbers.
6. B. 40. Counting by tens adds 10 each time, and 30 and 10 more is 40. If your child chose A (31), they added 1 instead of 10.
7. C. 15. Fifteen is 1 ten and 5 ones, so we write 1 and then 5. If your child chose B (51), they wrote the digits the wrong way round.
8. Compare 8 and 6: 8 is more than 6. So 8 is the greater number.
9. Compare 9 and 7: 7 is less than 9. So 7 is the smaller number.
10. A. 3. 13 is made of 1 ten and 3 ones, so the ones are 3. If your child chose B (1), they named the ten.
11. 10 and 4 more is 14. Counting back one from 20 gives 19.